

Tarea dección 0.1

$$7) \frac{1 + \cos 3t}{\sin 3t} + \frac{\sin 3t}{1 + \cos 3t} = 2 \csc 3t$$

$$\frac{(1 + \cos 3t)^2 + \sin^2 3t}{\sin 3t (1 + \cos 3t)} = 2 \csc 3t$$

$$\frac{1 + 2\cos 3t + \cos^2 3t + \sin^2 3t}{\sin 3t (1 + \cos 3t)} = 2 \csc 3t$$

$$\frac{1 + 2\cos 3t + 1}{\sin 3t (1 + \cos 3t)} = 2 \csc 3t$$

$$\frac{2(1 + \cos 3t)}{\sin 3t (1 + \cos 3t)} = 2 \csc 3t$$

$$\frac{2}{\sin 3t} = 2 \csc 3t$$

$$2 \csc 3t = 2 \csc 3t$$

$$9) \frac{1}{1 - \cos y} + \frac{1}{1 + \cos y} = 2 \csc^2 y$$

$$\frac{1 + \cos y + 1 - \cos y}{1 - \cos^2 y} = 2 \csc^2 y$$

$$\frac{2}{1 - \cos^2 y} = 2 \csc^2 y$$

$$\frac{2}{\sin^2 y} = 2 \csc^2 y$$

$$2 \csc^2 y = 2 \csc^2 y$$

$$17) \frac{\tan^2 X}{\sec X + 1} = \frac{1 - \cos X}{\cos X}$$

$$\frac{\frac{\sin^2 X}{\cos^2 X}}{\frac{1}{\cos X}} = \frac{1 - \cos X}{\cos X}$$

$$\frac{\sin^2 X \cos X}{\cos^2 X} = \frac{1 - \cos X}{\cos X}$$

$$\frac{\sin^2 X}{\cos X} = \frac{\sin^2 X}{\cos X}$$

$$21) \sin^4 r - \cos^4 r = \sin^2 r - \cos^2 r$$

$$(\sin^2 r - \cos^2 r)(\sin^2 r + \cos^2 r) = \sin^2 r - \cos^2 r$$

$$(\sin^2 r - \cos^2 r)(1) = \sin^2 r - \cos^2 r$$

$$\sin^2 r - \cos^2 r = \sin^2 r - \cos^2 r$$

$$25) (\sec t + \tan t)^2 = \frac{1 + \sec t}{1 - \sec t}$$

$$\left(\frac{1}{\cos t} + \frac{\sec t}{\cos t} \right)^2 = \frac{1 + \sec t}{1 - \sec t}$$

$$\left(\frac{1 + \sec t}{\cos t} \right)^2 = \frac{1 + \sec t}{1 - \sec t}$$

$$\frac{(1 + \sec t)^2}{1 - \sec^2 t} = \frac{1 + \sec t}{1 - \sec t}$$

$$\frac{1+\sec t (1+\sec t)}{(1+\sec t)(1-\sec t)} = \frac{1+\sec t}{1-\sec t}$$

$$\frac{1+\sec t}{1-\sec t} = \frac{1+\sec t}{1-\sec t}$$

$$33) \frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta - \sec \alpha \sec \beta} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta - \sec \alpha \sec \beta} = \frac{\frac{\sec \alpha}{\cos \alpha} + \frac{\sec \beta}{\cos \beta}}{1 - \frac{\sec \alpha}{\cos \alpha} \left(\frac{\sec \beta}{\cos \beta} \right)}$$

$$\frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta - \sec \alpha \sec \beta} = \frac{\frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta - \sec \alpha \sec \beta}{\cos \alpha \cos \beta}}$$

$$\frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta - \sec \alpha \sec \beta} = \frac{\sec \alpha \cos \beta + \cos \alpha \sec \beta}{\cos \alpha \cos \beta - \sec \alpha \sec \beta}$$

$$34) \sec \theta + \csc \theta - \cos \theta - \sin \theta = \sec \theta \tan \theta + \csc \theta \cot \theta$$

$$\frac{1}{\cos \theta} + \frac{1}{\sin \theta} - \cos \theta - \sin \theta = \sec \theta \left(\frac{\sin \theta}{\cos \theta} \right) + \csc \theta \left(\frac{\cos \theta}{\sin \theta} \right)$$

$$\frac{\sec \theta + \csc \theta - \cos \theta - \sin \theta}{\cos \theta \sin \theta} = \frac{\sec^2 \theta}{\cos \theta} + \frac{\csc^2 \theta}{\sin \theta}$$

$$\frac{\sec \theta (1 - \cos^2 \theta) + \csc \theta - \sin^2 \theta \csc \theta}{\cos \theta \sin \theta} = \frac{\sec^2 \theta + \csc^2 \theta}{\cos \theta \sin \theta}$$

$$\frac{\text{Sen}\theta \text{Sen}^2\theta + \cos\theta - \text{Sen}^2\theta \cos\theta}{\cos\theta \text{Sen}\theta} = \frac{\text{Sen}^2\theta + \cos^3\theta}{\cos\theta \text{Sen}\theta}$$

$$\frac{\text{Sen}^2\theta + \cos\theta(1 - \text{Sen}^2\theta)}{\cos\theta \text{Sen}\theta} = \frac{\text{Sen}^2\theta + \cos^3\theta}{\cos\theta \text{Sen}\theta}$$

$$\frac{\text{Sen}^2\theta + \cos\theta \cos^2\theta}{\cos\theta \text{Sen}\theta} = \frac{\text{Sen}^2\theta + \cos^3\theta}{\cos\theta \text{Sen}\theta}$$

$$\frac{\text{Sen}^2\theta + \cos^3\theta}{\cos\theta \text{Sen}\theta} = \frac{\text{Sen}^2\theta + \cos^3\theta}{\cos\theta \text{Sen}\theta}$$

$$43) \frac{\cot(-t) + \tan(-t)}{\cot t} = -\sec^2 t$$

$$\frac{\frac{\cos(-t)}{\text{Sen}(-t)} + \frac{\text{Sen}(-t)}{\cos(-t)}}{\cot t} = -\sec^2 t$$

$$\frac{\frac{\cos^2(-t) + \text{Sen}^2(-t)}{\text{Sen}(-t)\cos(-t)}}{\frac{\cos t}{\text{Sen} t}} = -\sec^2 t$$

$$\frac{1}{\frac{\text{Sen}(-t)\cos(-t)}{\cos t \text{Sen} t}} = -\sec^2 t$$

$$\frac{\text{Sen} t}{\text{Sen}(-t)\cos(-t)\cos t} = -\sec^2 t$$

$$-\frac{\sec t}{\sec t \cos t \cos t} = -\sec^2 t$$

$$\frac{-1}{\cos^2 t} = -\sec^2 t$$

$$-\sec^2 t = -\sec^2 t //$$

$$45) \log_{10} \tan t = \tan t$$

$$(1) \tan t = \tan t$$

$$\tan t = \tan t //$$

$$49) \ln |\sec \theta + \tan \theta| = -(\ln |\sec \theta - \tan \theta|)$$

$$\ln \left| \frac{(\sec \theta + \tan \theta)(\sec \theta - \tan \theta)}{\sec \theta - \tan \theta} \right| = -(\ln |\sec \theta - \tan \theta|)$$

$$\ln \left| \frac{\sec^2 \theta - \tan^2 \theta}{\sec \theta - \tan \theta} \right| = -(\ln |\sec \theta - \tan \theta|)$$

$$\ln \left| \frac{1}{\sec \theta - \tan \theta} \right| = -(\ln |\sec \theta - \tan \theta|)$$

$$\ln |1| - (\ln |\sec \theta - \tan \theta|) = -(\ln |\sec \theta - \tan \theta|)$$

$$-(\ln |\sec \theta - \tan \theta|) = -(\ln |\sec \theta - \tan \theta|) //$$